

Shuangyan Deng
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Personal Website: <https://jennie-deng.github.io/CV/>

Dear Goldman Sachs Engineering Recruiting Team,

I am writing to apply for the 2027 APEJ Singapore Engineering New Analyst Program at Goldman Sachs. I am currently a Ph.D. student in Statistics at the University of Auckland, with research focused on multimodal financial reasoning, financial artificial intelligence, and machine learning methods for financial decision-making. With a background spanning financial mathematics, banking data engineering, and AI research, I am excited by the opportunity to contribute to engineering teams that build scalable systems and apply technology to complex financial problems.

My academic and research work sits closely at the intersection of finance, data, and intelligent systems. I have developed benchmarks and methods for advanced financial reasoning across textual, numerical, tabular, visual, and multimodal evidence. My recent publications include work on multimodal financial reasoning, cross-model error reflection, and multi-agent evidence acquisition, including a Best Paper Award at ACM ICAIF 2025 for FinMR. This experience has strengthened my ability to reason rigorously about financial information, design evaluation frameworks, and translate complex research questions into practical computational solutions.

Before beginning my Ph.D., I worked as a Data Developer at Industrial Bank (Shanghai) Co., Ltd, where I built real-time data platforms for wealth management product sales and marketing operations. I helped design a unified data architecture integrating online and offline sales channels, implemented microservice-based systems using Spring Boot, ClickHouse, Redis, and Kafka, and supported low-latency querying and real-time operational visualization across multiple product lines. These projects gave me hands-on experience with scalable backend systems, financial data pipelines, cross-functional collaboration, and the practical demands of production-facing technology in a banking environment.

Goldman Sachs' Engineering division appeals to me because it combines technical depth with direct impact on global financial markets. I am particularly drawn to the firm's focus on scalable software, low-latency infrastructure, data-driven decision-making, and the use of machine learning alongside financial engineering. My visiting Ph.D. experience at Nanyang Technological University in Singapore has also strengthened my interest in contributing to Singapore's financial technology ecosystem. I would be grateful for the opportunity to bring my quantitative background, engineering experience, and research perspective to Goldman Sachs.

Thank you for your time and consideration. I look forward to the opportunity to discuss how my experience and interests align with the Engineering New Analyst Program.

Sincerely,

Shuangyan Deng